1. Introduction

This project is about designing and implementing an enterprise network using Cisco Packet Tracer. The network simulates a company with multiple departments such as HR, Finance, IT, and Admin. This project provides practical understanding of enterprise network design and implementation. It demonstrates how different networking technologies are integrated to ensure efficient communication and security within an organization.

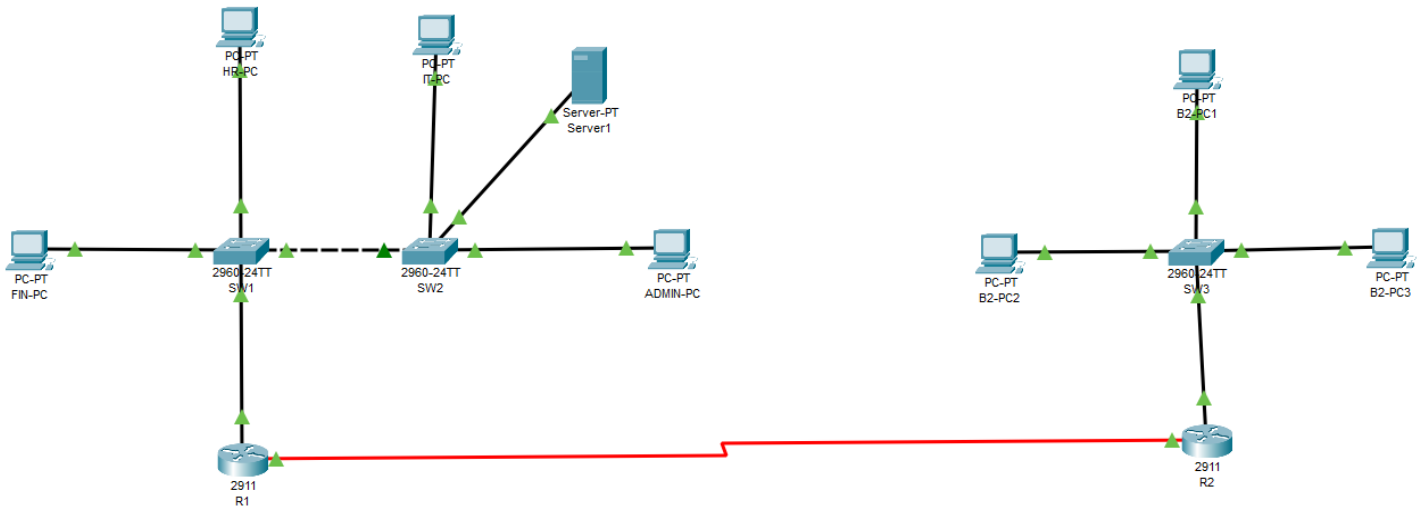
2. Objectives

- Create VLANs
- Configure inter-VLAN routing
- Connect branches using WAN
- Configure DHCP, DNS, Web
- Apply ACL security
- Test connectivity

3. Scope of the Project

This project covers VLANs, routing, WAN connection, and services for a basic enterprise network.

4. Network Topology



5. WAN Configuration

R1 and R2 Interface Configuration

R1:

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  192.168.10.1    YES manual up                    up
GigabitEthernet0/0.20  192.168.20.1    YES manual up                    up
GigabitEthernet0/0.30  192.168.30.1    YES manual up                    up
GigabitEthernet0/0.40  192.168.40.1    YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Serial0/3/0            10.0.0.1        YES manual up                    up
Serial0/3/1            unassigned      YES unset  down                  down
Vlan1                  unassigned      YES unset  administratively down down
R1#
```

R1 interfaces are up and correctly configured with LAN and WAN IP addresses.

R2:

```
R2>enable
R2#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.50.1    YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Serial0/3/0            10.0.0.2        YES manual up                    up
Serial0/3/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R2#
```

R2 interfaces are up and correctly configured with LAN and WAN IP addresses.

6. VLAN Configuration

VLAN Configuration on Switch

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Gig0/1, Gig0/2
10   HR                               active    Fa0/2
20   FINANCE                          active    Fa0/3
30   IT                               active
40   ADMIN                            active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Gig0/1, Gig0/2
10   VLAN0010                         active
20   VLAN0020                         active
30   VLAN0030                         active    Fa0/2
40   VLAN0040                         active    Fa0/3, Fa0/4
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

7. Trunk Configuration

SW1:

```
Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1
Fa0/24      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005
Fa0/24      1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1,10,20,30,40
Fa0/24      1,10,20,30,40

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,10,20,30,40
Fa0/24      1,10,20,30,40
```

SW2

```
Switch#show interface trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/24      on           802.1q         trunking      1

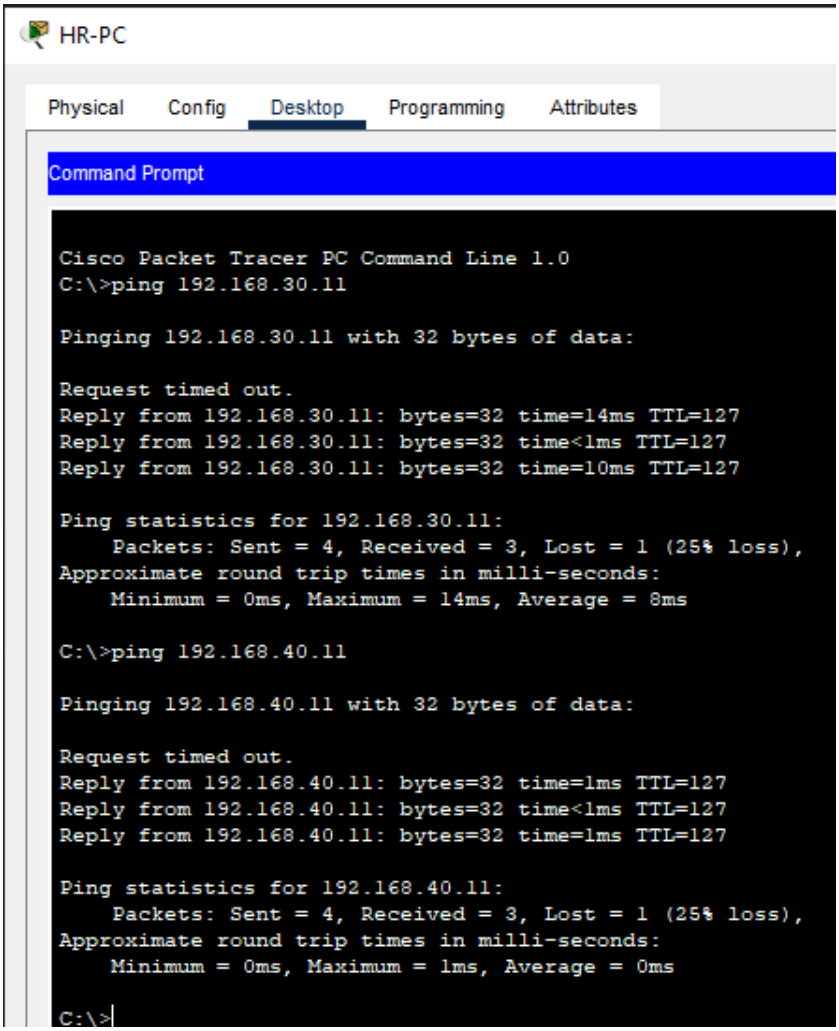
Port        Vlans allowed on trunk
Fa0/24      1-1005

Port        Vlans allowed and active in management domain
Fa0/24      1,10,20,30,40

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/24      1,10,20,30,40
```

8. Inter-VLAN Routing

Inter-VLAN Routing Test:



This test shows that devices in different VLANs can communicate through the router.

9. IP Addressing Plan

HR: 192.168.10.0/24

Finance: 192.168.20.0/24

IT: 192.168.30.0/24

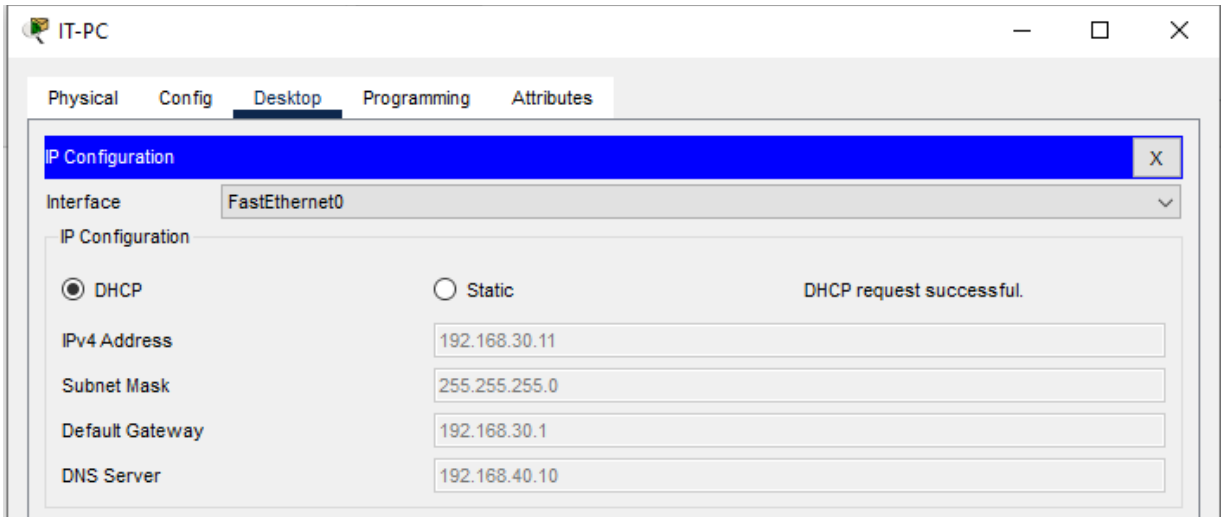
Admin: 192.168.40.0/24

Branch2: 192.168.50.0/24

10. DHCP Configuration

DHCP Configuration and IP Assignment:

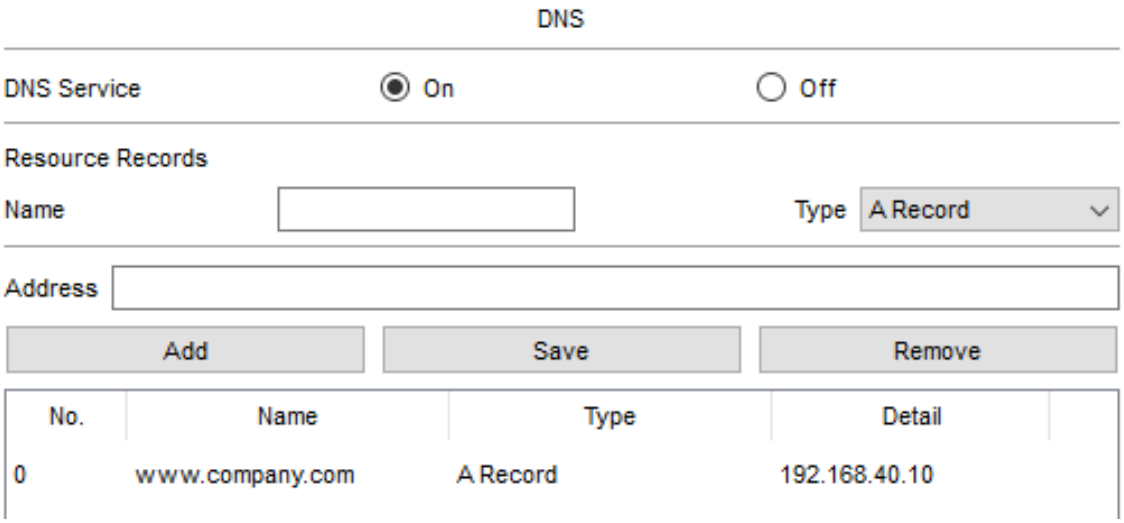
```
R1>enable
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.10
R1(config)#ip dhcp excluded-address 192.168.20.1 192.168.20.10
R1(config)#ip dhcp excluded-address 192.168.30.1 192.168.30.10
R1(config)#ip dhcp excluded-address 192.168.40.1 192.168.40.10
R1(config)#ip dhcp pool HR
R1(dhcp-config)#network 192.168.10.0 255.255.255.0
R1(dhcp-config)#default-router 192.168.10.1
R1(dhcp-config)#dns-server 192.168.40.10
R1(dhcp-config)#ip dhcp pool FINANCE
R1(dhcp-config)#network 192.168.20.0 255.255.255.0
R1(dhcp-config)#default-router 192.168.20.1
R1(dhcp-config)#dns-server 192.168.40.10
R1(dhcp-config)#ip dhcp pool IT
R1(dhcp-config)#network 192.168.30.0 255.255.255.0
R1(dhcp-config)#default-router 192.168.30.1
R1(dhcp-config)#dns-server 192.168.40.10
R1(dhcp-config)#ip dhcp pool ADMIN
R1(dhcp-config)#network 192.168.40.0 255.255.255.0
R1(dhcp-config)#default-router 192.168.40.1
R1(dhcp-config)#dns-server 192.168.40.10
R1(dhcp-config)#end
R1#
```



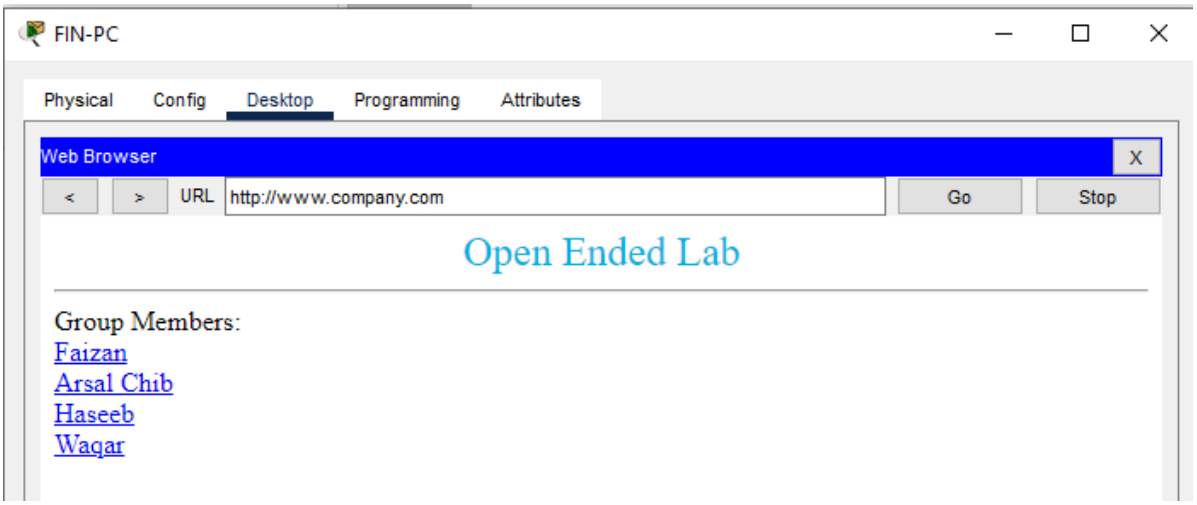
This shows that IP addresses are assigned automatically using DHCP.

11. DNS and Web Server

DNS and Web Server Configuration:



Web page Testing:



This shows that DNS is resolving domain name and web server is working correctly.

12. ACL

ACL Blocking HR to Finance Network

```
C:\>ping 192.168.20.11

Pinging 192.168.20.11 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.20.11:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Traffic is allowed for other networks

```
C:\>ping 192.168.30.11

Pinging 192.168.30.11 with 32 bytes of data:

Reply from 192.168.30.11: bytes=32 time<1ms TTL=127
Reply from 192.168.30.11: bytes=32 time<1ms TTL=127
Reply from 192.168.30.11: bytes=32 time=10ms TTL=127
Reply from 192.168.30.11: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.30.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>ping 192.168.40.10

Pinging 192.168.40.10 with 32 bytes of data:

Reply from 192.168.40.10: bytes=32 time<1ms TTL=127
Reply from 192.168.40.10: bytes=32 time=10ms TTL=127
Reply from 192.168.40.10: bytes=32 time<1ms TTL=127
Reply from 192.168.40.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.40.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>
```

13. Routing Configuration

Routing Table using RIP Protocol

```
R1>enable
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

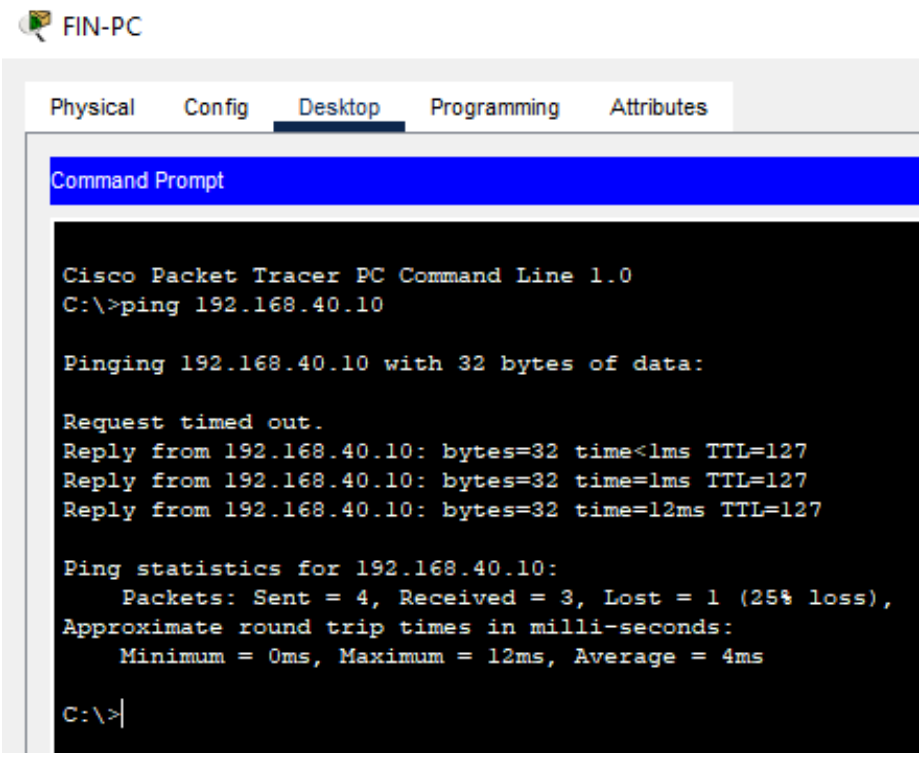
     10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/3/0
L       10.0.0.1/32 is directly connected, Serial0/3/0
C       10.0.0.2/32 is directly connected, Serial0/3/0
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0.10
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0.20
     192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, GigabitEthernet0/0.30
L       192.168.30.1/32 is directly connected, GigabitEthernet0/0.30
     192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.40.0/24 is directly connected, GigabitEthernet0/0.40
L       192.168.40.1/32 is directly connected, GigabitEthernet0/0.40
     192.168.50.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.50.0/24 is directly connected, GigabitEthernet0/0
L       192.168.50.1/32 is directly connected, GigabitEthernet0/0

R2>enable
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/3/0
C       10.0.0.1/32 is directly connected, Serial0/3/0
L       10.0.0.2/32 is directly connected, Serial0/3/0
R    192.168.10.0/24 [120/1] via 10.0.0.1, 00:00:23, Serial0/3/0
R    192.168.20.0/24 [120/1] via 10.0.0.1, 00:00:23, Serial0/3/0
R    192.168.30.0/24 [120/1] via 10.0.0.1, 00:00:23, Serial0/3/0
R    192.168.40.0/24 [120/1] via 10.0.0.1, 00:00:23, Serial0/3/0
     192.168.50.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.50.0/24 is directly connected, GigabitEthernet0/0
L       192.168.50.1/32 is directly connected, GigabitEthernet0/0
```

14. Testing and Results



HR PC Successful Communication

```
C:\>ping 192.168.40.10

Pinging 192.168.40.10 with 32 bytes of data:

Reply from 192.168.40.10: bytes=32 time<1ms TTL=127
Reply from 192.168.40.10: bytes=32 time=1ms TTL=127
Reply from 192.168.40.10: bytes=32 time<1ms TTL=127
Reply from 192.168.40.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.40.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Branch 2 Connectivity Test:

B2-PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ping 10.0.0.1

Pinging 10.0.0.1 with 32 bytes of data:

Reply from 10.0.0.1: bytes=32 time=1ms TTL=254
Reply from 10.0.0.1: bytes=32 time=1ms TTL=254
Reply from 10.0.0.1: bytes=32 time=2ms TTL=254
Reply from 10.0.0.1: bytes=32 time=1ms TTL=254

Ping statistics for 10.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>ping 192.168.40.10

Pinging 192.168.40.10 with 32 bytes of data:

Reply from 192.168.40.10: bytes=32 time=1ms TTL=126
Reply from 192.168.40.10: bytes=32 time=1ms TTL=126
Reply from 192.168.40.10: bytes=32 time=1ms TTL=126
Reply from 192.168.40.10: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.40.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>
```

B2-PC2

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.40.10

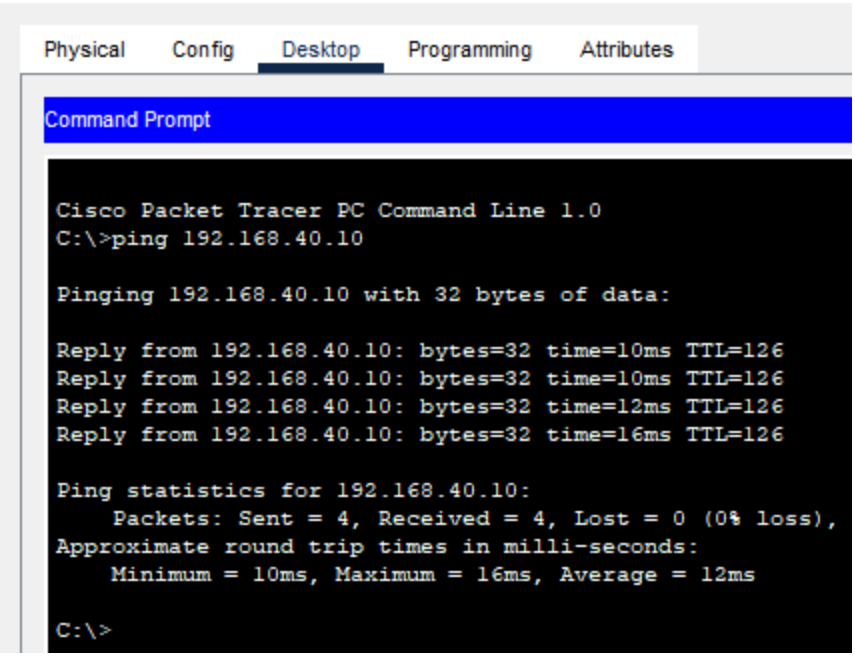
Pinging 192.168.40.10 with 32 bytes of data:

Reply from 192.168.40.10: bytes=32 time<1ms TTL=126
Reply from 192.168.40.10: bytes=32 time=2ms TTL=126
Reply from 192.168.40.10: bytes=32 time=10ms TTL=126
Reply from 192.168.40.10: bytes=32 time=2ms TTL=126

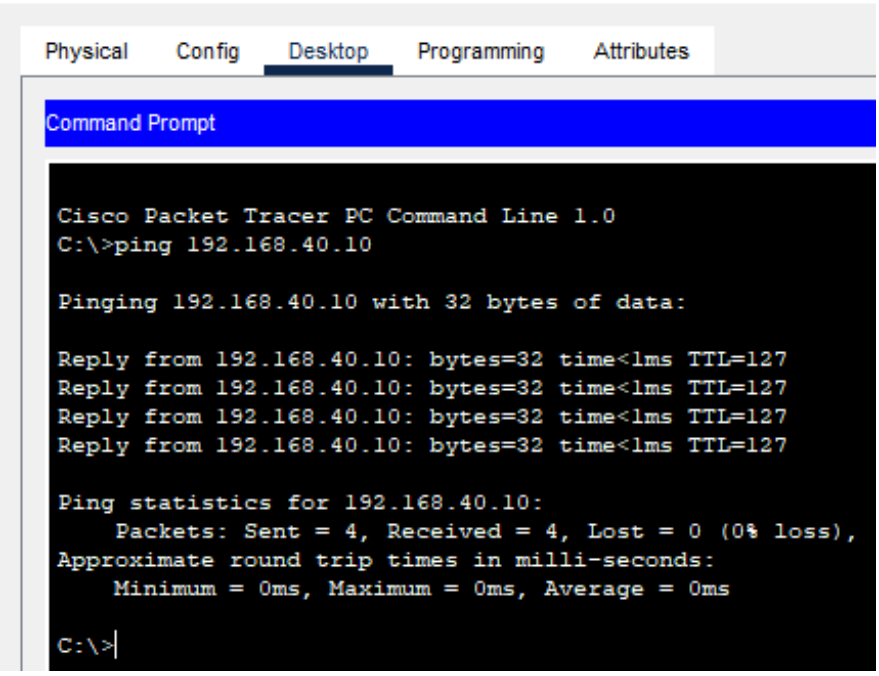
Ping statistics for 192.168.40.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 3ms

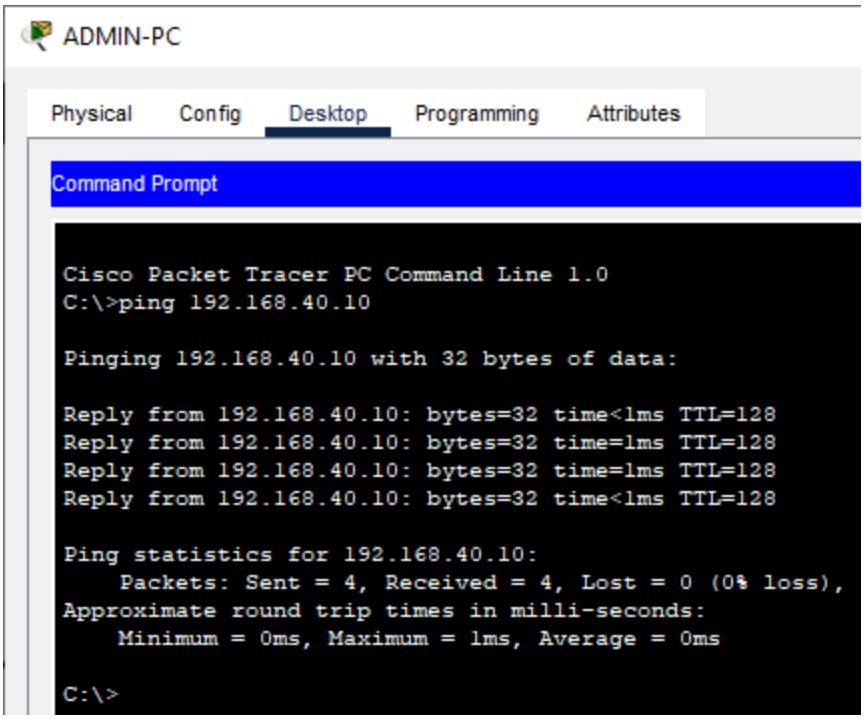
C:\>
```

B2-PC3


Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.40.10
Pinging 192.168.40.10 with 32 bytes of data:
Reply from 192.168.40.10: bytes=32 time=10ms TTL=126
Reply from 192.168.40.10: bytes=32 time=10ms TTL=126
Reply from 192.168.40.10: bytes=32 time=12ms TTL=126
Reply from 192.168.40.10: bytes=32 time=16ms TTL=126
Ping statistics for 192.168.40.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 10ms, Maximum = 16ms, Average = 12ms
C:\>

IT-PC


Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.40.10
Pinging 192.168.40.10 with 32 bytes of data:
Reply from 192.168.40.10: bytes=32 time<1ms TTL=127
Reply from 192.168.40.10: bytes=32 time<1ms TTL=127
Reply from 192.168.40.10: bytes=32 time<1ms TTL=127
Reply from 192.168.40.10: bytes=32 time<1ms TTL=127
Ping statistics for 192.168.40.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>



15. Issues Faced

During the implementation of this project, several issues were faced:

- Serial link was not working due to incorrect interface selection.
- CHAP authentication failed because of hostname mismatch between routers.
- Some PCs were not getting IP addresses due to DHCP configuration errors.
- Inter-VLAN communication initially failed due to trunk misconfiguration.
- Routing was not working properly because RIP networks were missing.
- First ping requests sometimes failed due to ARP delay.

All these issues were resolved by carefully checking configurations and testing step by step.

16. Networking Concepts Used

VLAN: Separates departments into different networks (HR, Finance, IT, Admin).

Trunking: Allows multiple VLAN traffic on one link using 802.1Q.

Inter-VLAN Routing: Router allows communication between VLANs.

WAN: Serial link connects two branches using PPP.

RIP v2: Routing protocol used between R1 and R2.

DHCP: Automatically assigns IP addresses.

DNS: Converts domain name to IP address.

Web Server: Hosts internal company website.

ACL: Blocks HR from accessing Finance network.

17. Network Workflow

HR to IT: HR-PC → Switch → Router → Switch → IT-PC.

Branch 2 to Server: B2-PC → R2 → WAN → R1 → VLAN → Server.

18. Advantages of the Network

- Improved security using VLANs and ACL.
- Easy network management.
- Efficient communication between departments.
- Automatic IP assignment using DHCP.
- Centralized services like DNS and Web.

19. Conclusion

In this project, an enterprise network was successfully designed and implemented using Cisco Packet Tracer. VLANs were used to separate departments, and inter-VLAN routing allowed communication between them. WAN connectivity was established between two branches using PPP, and RIP v2 was used for routing. DHCP, DNS, and Web services were configured and tested successfully. ACL was applied to enhance network security. All network components were tested, and the results confirmed that the network is working correctly. This project demonstrates a complete and functional enterprise network design.